

Week 01: Getting Started

CSU, Chico Math 351

2026-01-21

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R and Rstudio

R is a statistical programming language, while RStudio is the framework through which we will use R. You will need to install both for this course using the instructions on the next slide.

- ▶ **NOTE:** If you cannot install programs on your device, you will want to make an account at Posit Cloud to use R and RStudio in the course. [Posit Cloud](#)
- ▶ **NOTE:** A free Posit Cloud account should give about 25 hours/month, which may not be sufficient for this course. A Cloud Student account is \$5/month for 75 hours/month, which should be enough. If this will not work, other options include using the physical campus computers, Wildcat Virtual Lab, or Tech Lending programs through the library for in-class work.

Please only use the Cloud option if you have no other options possible. Your own installation ensures you have greater control of your files, and that you're not dependent on a stable internet connection.

Download/Install R

► Download

- Windows: [R-4.5.2](#)
- macOS: [R-4.5.2](#) (Check under "Latest release")
 - If you get a message about needing X11 or XQuartz, that can be downloaded at the same link.
 - Save the file, but do not open or run it.

► Install

- Install R by double clicking on the downloaded file and following the prompts.
- Stick with default settings. Remove any desktop shortcuts and eject any disk images (macOS)

► Setup Tutorials

[Windows](#)

[Mac](#)

Download/Install RStudio

► Download

- Download the current version here under **“2. Install RStudio”**: [RStudio Desktop](#)
- The download page will automatically recognize your OS, but note whether it supports the current version. You may need a [previous version](#) in some cases.
- Save the file, but do not open or run it.

► Install

- Windows: Double click on the downloaded file to run the installer program.
- Mac: Double click on the downloaded file, then drag the R Studio Icon into your Applications folder.

RStudio Preferences

A few quick setting changes will save you some trouble down the line:

- ▶ Open RStudio. In the file menu, go to *Tools*, then *Global Options*
- ▶ Uncheck “Restore .RData into workspace at startup”
- ▶ Where it says “Save workspace to .RData on exit:” select “Never”
- ▶ Click “Apply” first to save the changes, then “OK” to close the window.

With this change, you won't accidentally carry over work from previous sessions.

Additionally, you can restart R by going to the file menu, clicking *Session*, then “Restart R”.

Adding $\text{\LaTeX}$

To handle math notation in R, we'll need the typesetting program $\text{\LaTeX}$. A lightweight version is found in the R package `tinytex` that will allow us to write math equations and generate PDFs for homework assignments.

- ▶ In your RStudio console pane (typically the bottom-left window), type the following and press Enter:

```
install.packages("tinytex")
```

- ▶ In the console pane, type the following and press Enter:

```
tinytex::install_tinytex()
```


Additional Packages

Besides `tinytex`, other contributed packages exist containing functions that will help us with much of the statistics we will learn. You can install them by typing:

```
install.packages("package name")
```

replacing “package name” with the appropriate name. You’ll want to install the following in order:

- ▶ `openintro`
- ▶ `devtools`

Downloading Materials

You'll want a consistent file and folder structure to best organize your work for this course. Be sure to stick with a single naming convention for an easier time navigating filepaths for some assignments. Examples:

- ▶ ALL CAPS: MATH350
- ▶ no caps: math350
- ▶ snake_case: math_350
- ▶ CamelCase: Math350
- ▶ NOTE: Avoid spaces in folder and file names!

Please take a moment now to create a new folder named "MATH350" with one of the above naming conventions.

Using Your Folder

- ▶ Don't let files live in your Downloads folder!
- ▶ When downloading class materials and files, right click and "Save as" or "Save target as" and **actively choose** where to download this file.
- ▶ Also do not open any files from your browser window after downloading.

Please take a moment now to save the course Syllabus to your new folder using the following link: [Syllabus](#).

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Adapted from notes by Dr. Donatello and from the [MATH 130 website](#)